

Modern Methods of Data Analysis

Project Report

Project: Cyberbullying Tweets Classification

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1. Introduction

1.1. Subject area

The widespread use of social media platforms such as Twitter has brought unprecedented opportunities for communication and self-expression, but it has also facilitated the rapid proliferation of harmful content, including cyberbullying. Automated detection of cyberbullying is a critical application of natural language processing (NLP) for online safety. However, many existing approaches treat cyberbullying detection as a binary classification task (bullying vs. non-bullying), which fails to capture the nuanced nature of harmful content. A fine-grained typology — distinguishing between different types of cyberbullying such as those targeting age, ethnicity, gender, or religion — can significantly improve downstream content moderation pipelines, where different categories may require distinct policy responses. The goal of this project is to build a supervised multi-class single-label text classifier that, given the raw text of a tweet, predicts whether it constitutes cyberbullying and, if so, identifies its specific type. The label space consists of six mutually exclusive classes: age, ethnicity, gender, religion, other_cyberbullying, and not_cyberbullying. This report documents the complete project implementation: (i) exploratory data analysis to understand data characteristics, (ii) preprocessing to clean and normalize the noisy tweet text, (iii) feature engineering to extract discriminative signals for subsequent classification, and (iv) training and evaluation on train/test data.

1.2. Task distribution

Gorelskii Roman - Preprocessing + Feature engineering + SHAP

Radaeva Karina - Linear and ensemble models analysis

Plohotnyuk Valeria - Transformer model analysis

2. Dataset summary

2.1. Initial preprocessing results

2.1.1. Source and high-level overview

The dataset is publicly available Cyberbullying Classification corpus published by Larxel on Kaggle (<https://www.kaggle.com/datasets/andrewmvd/cyberbullying-classification>), originally collected and described in Wang, Fu, Lu (2020). SOSNet: A Graph Convolutional Network Approach to Fine-Grained Cyberbullying Detection (IEEE Big Data 2020). The data consists

of mostly English tweets that were sampled and labelled into one of six categories of cyberbullying or non-cyberbullying.

The CSV file (cyberbullying_tweets.csv) contains 47,692 labelled tweets and two columns:

- tweet_text – raw textual content of a tweet (string).
- cyberbullying_type – class label (categorical, 6 levels).

There are no missing values in either column. The dataset is large enough to satisfy the assignment constraint of at least 50 objects with at least 7-10 (potentially engineered) features: although the raw schema has only one feature, vectorised text representations such as bag-of-words, TF-IDF n-grams, and tweet-level meta-features (length, hashtag/URL/mention counts, share of non-ASCII characters, etc.) easily supply hundreds of derived features, of which the ones used in this work are introduced in Section 3.

Property	Value
Number of tweets (rows)	47,692
Number of columns	2
Missing values	0 in tweet_text, 0 in cyberbullying_type
Exact duplicate rows	36
Duplicate tweet_text values	1,675
Number of classes	6
Unique tokens (regex tokenizer, lower-cased)	62,663

Table 1. Basic dataset statistics.

2.1.2. Class distribution

The label distribution is almost perfectly balanced – every class contains between 7,823 and 7,998 tweets, i.e. each class accounts for approximately 16.4–16.8% of the total. This is a deliberate design choice of the original dataset's authors and is very convenient for modelling: it removes the need for class re-weighting, oversampling or stratified loss functions to combat class imbalance, and it allows accuracy-based metrics to remain meaningful. We will, however, still report macro-averaged precision, recall and F1 to make per-class behaviour explicit, since some classes (in particular other_cyberbullying) are likely to be linguistically less coherent than others.

Class	Count	Share, %
religion	7,998	16.77
age	7,992	16.76
gender	7,973	16.72
ethnicity	7,961	16.69
not_cyberbullying	7,945	16.66
other_cyberbullying	7,823	16.40

Table 2. Class distribution.

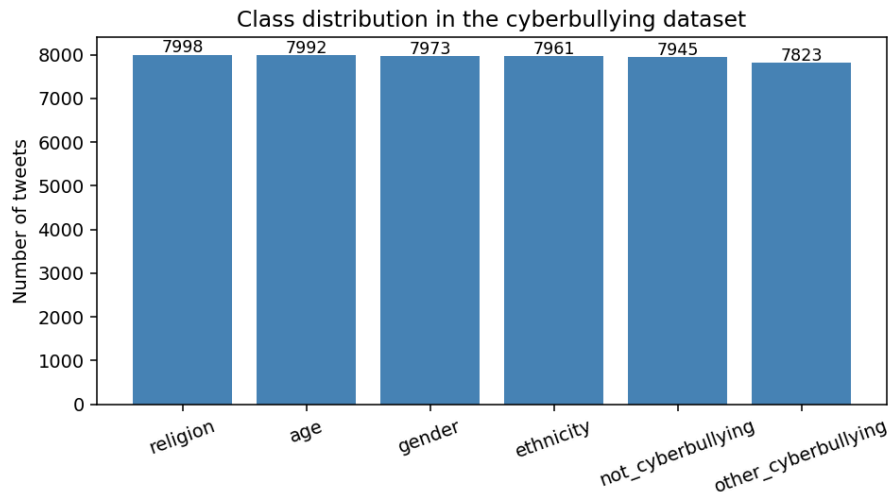


Figure 1. Class distribution of the cyberbullying dataset (almost balanced).

2.1.3. Tweet length

Two length statistics were computed on the raw text: number of characters and number of tokens (words + punctuation as defined by the regex tokenizer used in the notebook). Globally, tweets are short and have a strongly right-skewed length distribution, which is consistent with the platform's historical 140/280-character limit:

Statistic	Characters	Tokens
min	1	0
25%	78	13
median	124	20
75%	180	32
max	5,018	912
mean	136.3	23.7
std	85.2	15.4

Table 3. Length statistics over all 47,692 tweets.

The maximum length (5,018 characters / 912 tokens) is most likely an outlier – the platform limit makes such values impossible for legitimate tweets, so these records are most likely concatenations of multiple tweets, retweet chains, or noise introduced during data collection. They will be flagged for inspection and possibly truncated or removed at the preprocessing stage.

Length is far from class-independent: per-class averages span a factor of more than two.

Class	Char mean	Char median	Tokens mean	Tokens median	Tokens std
age	173.5	171	34.6	34	16.2
religion	198.0	212	36.5	39	14.3
ethnicity	139.3	126	27.0	24	15.6
gender	136.4	129	26.0	23	14.1
other_cyberbullying	85.7	83	16.0	16	17.1
not_cyberbullying	83.1	81	15.2	15	8.5

Table 4. Per-class length statistics.

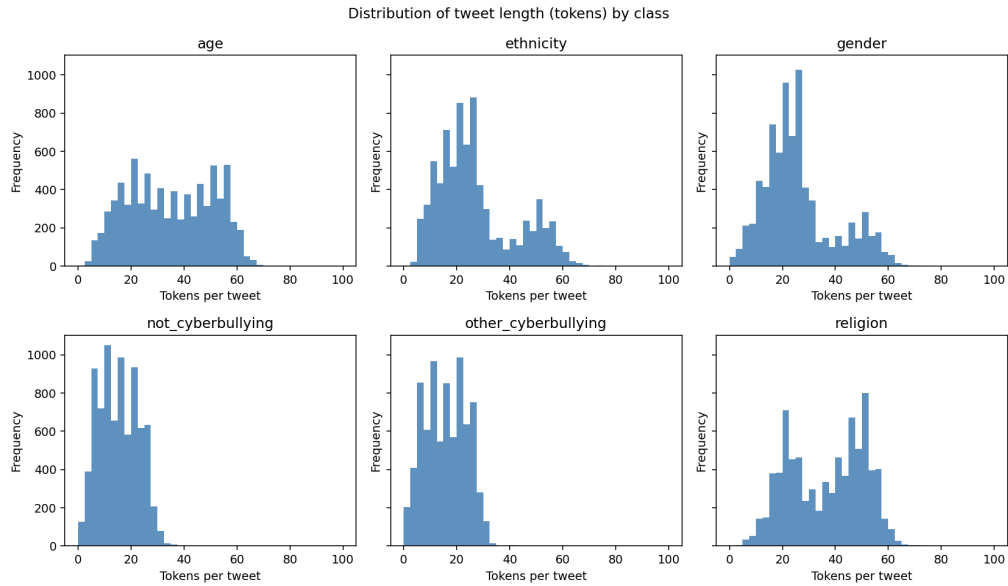


Figure 2. Distribution of tweet length (in tokens) per class.



Figure 3. Tweet length per class (boxplot, outliers hidden).

Two patterns are visible. First, religion and age tweets are systematically longer (median 39 and 34 tokens), which reflects the fact that tweets in these classes often describe personal narratives or quote longer arguments. Second, not_cyberbullying and other_cyberbullying are the shortest (median ~15–16 tokens) and have the heaviest right tails (the std of other_cyberbullying token count is 17.1, larger than its mean), which we attribute to the residual nature of the other_cyberbullying class – it appears to absorb tweets that did not fit any of the four targeted categories and is therefore more heterogeneous. Length alone is already a moderately discriminative feature and will be added to the feature set as a numerical meta-feature.

2.1.4. Twitter-specific tokens (URLs, mentions, hashtags)

Twitter syntax leaves easily identifiable artefacts in the text. Across the corpus, 9.0% of tweets contain at least one URL, 38.0% contain at least one @mention, and 14.6% contain at least one #hashtag. Their per-class behaviour is informative:

Class	Mean URLs	Mean @mentions	Mean #hashtags
not_cyberbullying	0.149	0.776	0.412
other_cyberbullying	0.177	0.861	0.208
gender	0.108	0.569	0.338
religion	0.070	0.518	0.225
ethnicity	0.044	0.616	0.140
age	0.026	0.065	0.092

Table 5. Mean number of URLs, @mentions and #hashtags per tweet, by class.

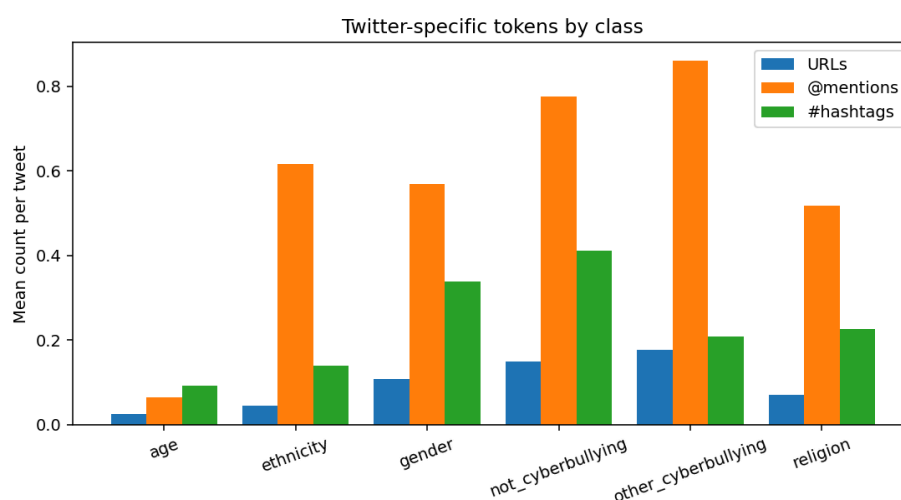


Figure 4. Mean number of Twitter-specific tokens per tweet, by class.

The non_cyberbullying and other_cyberbullying classes are by far the most "conversational" (more @mentions and URLs), whereas the age class is striking in the opposite direction – almost no @mentions (0.07 per tweet on average), which already hints that the data-collection query for this class targeted descriptive narratives ("in high school I was bullied ...") rather than direct exchanges. URL/mention/hashtag counts will be retained as engineered numerical features.

2.1.5. Vocabulary and most frequent words

The corpus was tokenised with the same regular expression that is used in the notebook (sequences of word characters with hyphens and apostrophes, plus single punctuation symbols), and tokens were lower-cased. The full vocabulary consists of 62,663 distinct tokens. Per class, the type/token ratios reveal that not_cyberbullying and other_cyberbullying are lexically the most diverse:

Class	Tokens (total)	Unique tokens	TTR
religion	292,105	19,691	0.067
age	276,667	16,025	0.058
ethnicity	214,595	18,017	0.084
gender	207,035	17,548	0.085

other_cyberbullying	125,072	16,031	0.128
not_cyberbullying	120,769	19,201	0.159

Table 6. Vocabulary size and type/token ratio per class.

The two most lexically diverse classes (not_cyberbullying with TTR=0.16 and other_cyberbullying with TTR=0.13) are also the smallest in total token count, confirming that these classes are linguistically more open. Religion and age, on the contrary, are dominated by a comparatively narrow vocabulary.

The top-15 frequent words per class are dominated by stopwords, punctuation and URL fragments (., the, a, to, i, http, t, co, ...), which is expected. Once class-agnostic tokens are filtered out, distinctive content terms appear immediately:

- age – high, school, bullied, me, was, my (top bigrams: high school, in high, i was, bullied me, school bully). The class is dominated by autobiographical narratives about school bullying.
- religion – muslims, muslim, are, you (top bigrams: islamic terrorism, christian woman, you are, a christian). Tweets centre on Islam and Christianity.
- gender – rape, gay, jokes, joke, sexist, feminazi (top bigrams: rape jokes, rape joke, gay jokes, not sexist, sexist but, i'm not). Heavy presence of trolling templates such as "I'm not sexist but ...".
- ethnicity – fuck, dumb, nigger, ass, black (top bigrams: dumb ass, ass nigger, fuck obama, dumb nigger). The class is dominated by racially-charged slurs and is by far the most lexically aggressive.
- not_cyberbullying – mkr, http, t, co (the show #MyKitchenRules dominates this class because of the way the dataset was sampled).
- other_cyberbullying – feminazi, blameonenotall, coon, plus a long tail of Twitter handles and hashtags; the class also concentrates ð characters from broken emoji encodings, which is a useful data-quality signal.

2.1.6. Language and data-quality observations

Although the dataset is described as English, the texts are not exclusively English. The notebook computes, for every tweet, an "English quota" (the share of regex tokens that belong to the NLTK English wordlist) and visualises its distribution. While the majority of tweets sit in the 0.5-1.0 range, there is a non-trivial mass with low English quota – tweets containing Portuguese, Spanish, Hindi/Urdu transliterations, Arabic, broken emoji encodings (sequences of ð and ã caused by encoding issues), and class-specific slang or hashtags. Two pieces of evidence support this:

- Across the whole corpus, 19.2% of tweets contain at least one non-ASCII character, which is a conservative proxy for non-English content (it ignores tweets that use the ASCII range to write transliterated non-English text).
- The class-specific top words include Portuguese forms ('é', 'não', 'por', 'você', 'mais', 'sofre') in not_cyberbullying and Hindi/Urdu/Indic-script related terms ('kashmir', 'pakistani', 'modi', 'kashmiri', 'hai') in religion.

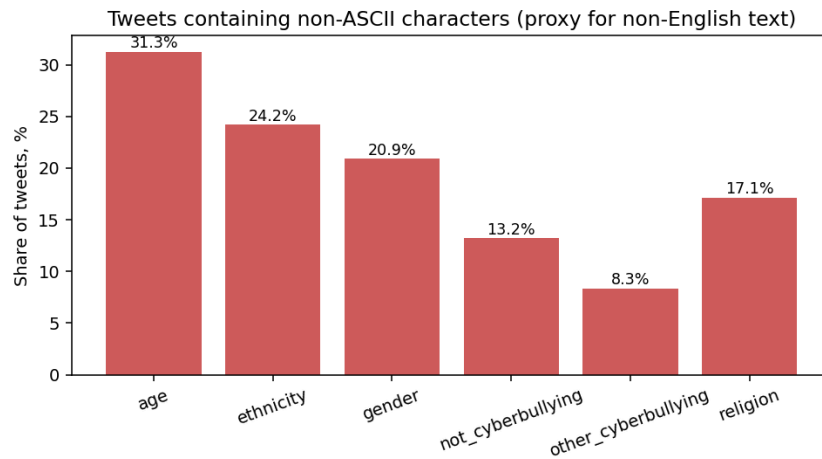


Figure 5. Share of tweets containing non-ASCII characters, by class.

Per-class non-ASCII shares are noticeably uneven (31% in age, 24% in ethnicity, 8% in other_cyberbullying), so any decision about non-English data has class-dependent consequences.

An obvious option would be to filter out tweets whose English quota falls below some threshold and analyse only the English subset. This was tested on the notebook level and rejected for the EDA stage: when the non-English tweets are dropped, the discriminative power of class-specific tokens (the words that occur only in one cyberbullying class and nowhere else) drops noticeably. In other words, a non-negligible share of the most class-characteristic vocabulary is in fact non-English or transliterated, and removing it throws away signal. We therefore keep all tweets at this stage and treat multilinguality as one of the open questions for the modelling phase, where the choice of preprocessing pipeline (e.g. NLTK English wordlist vs. language-id + per-language lemmatisation, or a multilingual embedding such as XLM-R) will be empirically compared.

Other data-quality issues identified during EDA:

- 1,675 tweets have a duplicated tweet_text value (3.5% of the corpus). 36 of those are exact duplicates of full rows. Some of the duplicates appear with different labels and will need to be deduplicated carefully (we will keep the row with the label that matches the majority vote within the duplicate group).
- A handful of tweets are extreme length outliers (up to 5,018 characters / 912 tokens), well above the platform limit; these will be inspected and likely truncated or excluded.
- Several tweets contain mojibake (broken emoji bytes such as Ď, ã) coming from a non-UTF8 source; these will be normalised with ftfy or removed.
- The not_cyberbullying class is dominated by tweets about #MyKitchenRules ("mkr" is the 6th most frequent token there). This is a sampling artefact that may inflate apparent classification accuracy; we will check whether models pick up on the show-specific vocabulary by inspecting feature importances and by evaluating on hand-curated held-out subsets.
- URLs and @mentions, although they carry signal at the meta-feature level, will be replaced with stable placeholders (URL, USER) before tokenisation, to prevent the vectoriser from blowing up the vocabulary with unique IDs.

2.2. Updated preprocessing results

2.2.1. English quota

An important finding from the EDA is that not all tweets are in English, despite the dataset being described as English-dominant. To quantify this, an "English quota" was computed for each tweet: the proportion of tokens that appear in the NLTK English wordlist. The distribution histogram shows a significant number of tweets with English quota below 0.5. Many tweets contain non-English content (Portuguese, Spanish, Hindi/Urdu transliterations, Arabic, or broken emoji encodings). Based on this finding, tweets with an English quota below 0.8 (less than 80% English tokens) were filtered out for the core analysis. This ensures that subsequent modeling focuses primarily on English content (and that words not from NLTK English wordlist can still be present), though it comes at the cost of losing some class-specific signal from multilingual tweets.

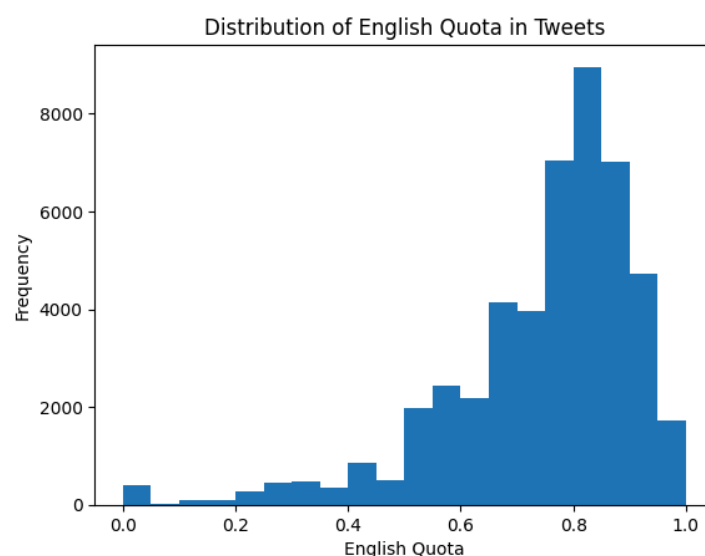


Figure 6. Distribution of English quota in tweets

2.2.2. Class distribution with English quota applied

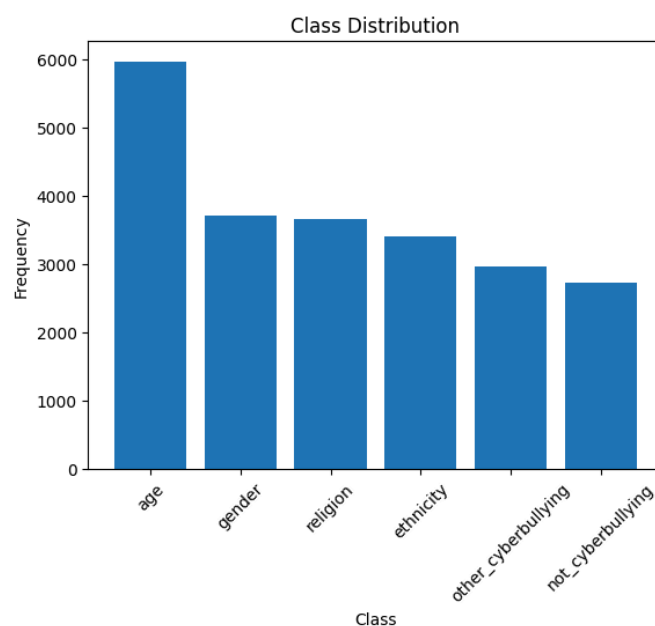


Figure 7. Class distribution after application of English quota.

The dataset after filtering is NOT balanced. The age class has approximately twice as many tweets (~6,000) as the smallest classes (not_cyberbullying ~2,900 and other_cyberbullying ~3,000). This is an important finding that will require handling class imbalance during the modeling phase.

2.2.3. Vocabulary analysis

The updated vocabulary analysis reveals 28,617 unique words across the corpus, with age and religion classes exhibiting the largest vocabularies (12,242 and 10,772 unique words respectively), while not_cyberbullying and other_cyberbullying have the smallest (6,843 and 6,924). Notably, 21,774 words (76% of all unique words) appear exclusively in cyberbullying classes, confirming distinct linguistic signals for harmful content. Gender-specific terms include "assault", "feminazi", and "slurs" reflecting patterns of sexual harassment. Religion class is characterized by "islamic", "christians", "terrorists", "hindu" and geopolitical terms like "india" and "pakistan." Age class centers on school bullying narratives with words such as "school", "bullied", "hated" and "grew". Other_cyberbullying contains many unique Twitter handles and usernames, indicating targeted harassment campaigns, while not_cyberbullying has very few unique words, suggesting its vocabulary largely overlaps with other classes. These distinctive lexical patterns confirm that linear classification models should perform well on this task.

3. Preprocessing

A custom TextPreprocessor class was implemented to handle the noise and irregularities present in the tweet corpus. The class is designed with configurable parameters to allow experimentation with different preprocessing strategies.

3.1. Preprocessing steps

- **Mojibake normalization** Broken Unicode sequences (e.g., \u0080, \u00e2), common in the dataset due to encoding issues, are repaired using the ftfy library.
- **Placeholder replacement** URLs are replaced with [URL], @mentions are replaced with [MENTION]. This prevents the vectorizer from creating unique tokens for each URL or username, which would bloat the vocabulary without adding a generalizable signal.
- **Lower-casing** All text is converted to lowercase to ensure case-insensitive matching and reduce vocabulary size.
- **Tokenization** Text is tokenized using a custom regex pattern, followed by NLTK's word_tokenize, which separates punctuation from words.
- **Stopword removal** Common English stopwords (from NLTK's stopword list) are removed to focus on content-bearing terms. Lemmatization or Stemming (configurable):
- **Lemmatization (WordNetLemmatizer)** Reduces words to their base dictionary form (e.g., "running" → "run")
- **Stemming (PorterStemmer)** Removes suffixes more aggressively (e.g., "running" → "run")
- **Duplicate token removal** Within each tweet, duplicate tokens are removed to reduce noise from repeated words or retweet artefacts.

- **Minimum word length filtering** Tokens shorter than a configurable minimum length (default 2 characters) are discarded.

3.2. Preprocessing configuration testing

To identify the optimal preprocessing strategy, five different configurations were tested on a random sample of 5,000 tweets. The configurations varied across lowercasing, stopwords removal, stemming vs. lemmatization, URL/mention replacement, mojibake normalization, duplicate removal, and minimum word length.

Configuration	Average tokens	Unique ratio	Empty tweets (%)
Config 1 (baseline, no normalization)	16.9	0.1683	0.0
Config 2 (+ normalization, replacement)	16.2	0.1313	0.1
Config 3 (+ stemming, deduplication)	14.9	0.1168	0.1
Config 4 (+ lemmatization)	15.0	0.1413	0.1
Config 5 (min_word_length=3)	14.0	0.1479	0.1

Selected configuration for the final pipeline:

- Lowercase
- Remove stopwords
- Use lemmatization
- Replace URLs and mentions
- Normalize mojibake
- Remove duplicates
- Minimum word length: 2

This configuration balances token diversity (unique ratio ≈ 0.14) while keeping the proportion of tweets reduced to empty strings below 0.1%.

4. Feature engineering

A FeatureEngineer class was implemented to extract two categories of features from the preprocessed tweets.

4.1. TF-IDF features

Term Frequency-Inverse Document Frequency (TF-IDF) features capture the importance of n-grams within each tweet relative to the entire corpus.

Parameters used:

- N-gram range: (1, 2) — unigrams and bigrams to capture both individual keywords and common two-word phrases identified in the EDA (e.g., high school, rape jokes, islamic terrorism)
- max_features: 30,000 — limit vocabulary size for computational efficiency
- min_df: 3 — ignore terms appearing in fewer than 3 documents
- max_df: 0.9 — ignore terms appearing in more than 90% of documents (to filter out overly common terms like punctuation)
- sublinear_tf: True — use $1+\log(\text{tf})$ to dampen the effect of very high frequencies

4.2. Meta features added

Based on the EDA the following meta-features were derived:

Feature	Description
token_count	Number of tokens after preprocessing
char_count	Total characters (sum of token lengths)
avg_token_len	Average token length ($\text{char_count} / \text{token_count}$)
url_count	Number of [URL] placeholders
mention_count	Number of [MENTION] placeholders
hashtag_count	Number of tokens starting with #
non_ascii_share	Proportion of characters with ASCII code > 127
punctuation_share	Proportion of punctuation characters (!?.!?:;,"')
exclamation_count	Number of exclamation marks (!)
English quota	Proportion of tokens in a tweet that appear in the NLTK English vocabulary. It is computed as the ratio of tokens recognized as English words to the total number of tokens after tokenization. Values close to 1 indicate predominantly English tweets, whereas lower values suggest the presence of non-English content, transliterated text, hashtags, usernames, or encoding artifacts.

4.3. Train/test split

The TF-IDF representation was generated using unigrams and bigrams with `max_features=30000`, `min_df=3`, and `max_df=0.9`. After fitting the vectorizer on the training data, the resulting vocabulary contained 12,765 features. These sparse TF-IDF features were horizontally combined with 10 scaled meta-features (including the English quota feature) using `scipy.sparse.hstack`, resulting in a final feature matrix with 12,775 features. The target labels (`cyberbullying_type`) were encoded numerically using `LabelEncoder` into six classes (0–5). The dataset was split using a stratified 60/20/20 train/validation/test strategy to preserve the class distribution. First, 20% of the data was reserved as a held-out test set. The remaining 80% was further divided into training and validation subsets, resulting in 12,662 training instances, 4,224 validation instances, and 4,224 test instances. The same data partitioning scheme was subsequently used for both classical machine learning models and Transformer-based models to ensure a fair comparison.

The use of stratified sampling ensures that each subset maintains approximately the same class proportions as the original dataset. This is particularly important given the class imbalance introduced after filtering tweets using the English quota threshold, allowing model performance to be evaluated consistently across all cyberbullying categories.

5. Linear and ensemble models

5.1 Model training configuration

Several classical machine learning models were evaluated on the TF-IDF and meta-feature representation described in Section 4. The goal was to establish a strong baseline and to compare different linear and ensemble approaches before introducing Transformer-based models.

First, three linear classifiers were trained directly on the sparse TF-IDF + meta-feature representation: Logistic Regression, Linear SVM, and SGD Linear SVM. In addition, a hard Voting Classifier combining these three linear models was trained.

- `Logistic Regression(C=4.0, max_iter=2000, solver="saga", n_jobs=-1, random_state=42, multi_class="multinomial")`
- `Linear SVM(C=1.0, random_state=42)`
- `SGD Linear SVM(loss="hinge", alpha=1e-5, max_iter=1000, tol=1e-3, random_state=42, n_jobs=-1)`
- `Voting Classifier(hard voting over Logistic Regression + Linear SVM + SGD Linear SVM, voting="hard", n_jobs=-1)`

Tree-based ensemble models were also considered. Since decision-tree methods are not well suited to very high-dimensional sparse TF-IDF matrices, the TF-IDF representation was first reduced using `TruncatedSVD` with 300 components. The reduced dense representation was then concatenated with the scaled meta-features and used to train `Random Forest`, `Extra Trees`, and `HistGradientBoosting` classifiers.

- `Random Forest(n_estimators=300, max_depth=None, min_samples_split=2, min_samples_leaf=1, random_state=42, n_jobs=-1, class_weight=None)`

- Extra Trees(n_estimators=300, max_depth=None, min_samples_split=2, min_samples_leaf=1, random_state=42, n_jobs=-1, class_weight=None)
- HistGradientBoosting(max_iter=200, learning_rate=0.08, max_leaf_nodes=31, random_state=42)

5.2 Methodology

These models were chosen to cover several strong classical baselines for sparse text classification. Logistic Regression, Linear SVM, and SGD Linear SVM are widely used for TF-IDF features because they scale well to high-dimensional sparse matrices and often perform very strongly on text data. Logistic Regression also provides interpretable coefficients, which are useful for feature-importance and SHAP analysis.

The Voting Classifier was included to test whether combining several linear models could improve robustness by reducing individual model errors. Random Forest, Extra Trees, and HistGradientBoosting were added as non-linear ensemble alternatives to check whether more flexible tree-based models could capture interactions between features.

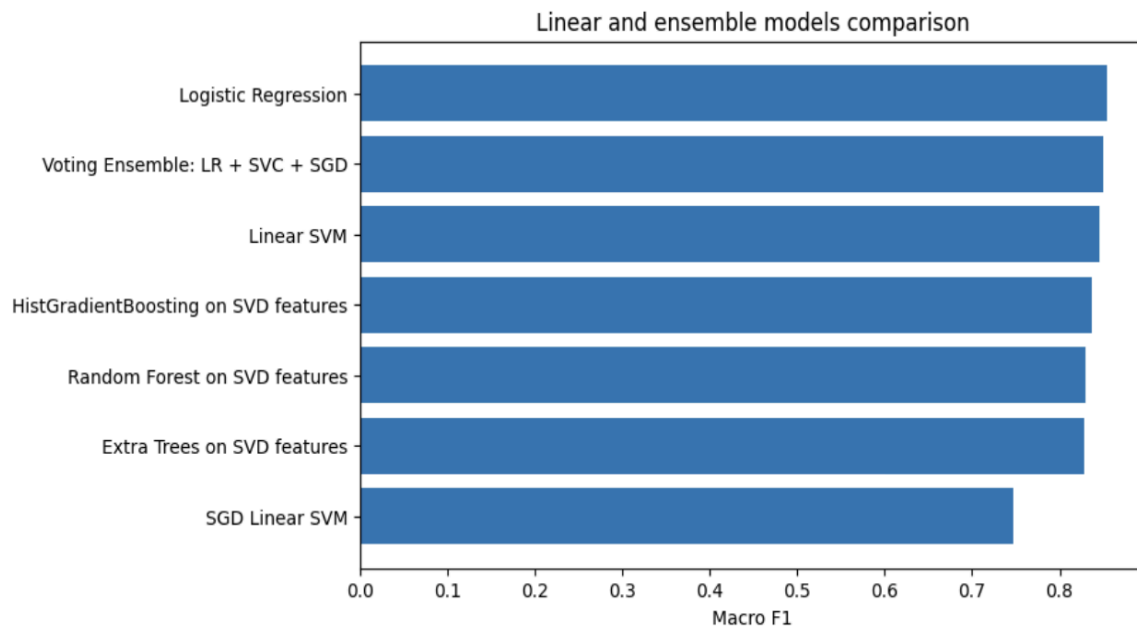
After applying the English quota threshold of 0.8, thereby retaining only tweets containing at least 80% English tokens, the combined feature matrix consisting of TF-IDF features and engineered meta-features was partitioned into training, validation, and test sets using stratified sampling to preserve class proportions. First, 20% of the data was reserved as a held-out test set. The remaining 80% was further split into training and validation sets, resulting in an overall 60/20/20 % train/validation/test split. The training set was used to fit the models, the validation set was used for model selection and comparison, and the held-out test set was reserved exclusively for the final evaluation of the best selected model. The same data split was used for both the classical and Transformer-based approaches to ensure a fair comparison. Model performance was assessed using accuracy together with macro-averaged precision, recall, and F1-score to account for the class imbalance introduced by the English quota filtering. The best model among classical approaches was chosen based on the highest macro F1-score.

5.3 Linear and ensemble models comparison results

The overall performance of the evaluated models on the validation set is summarized in the table:

Model	Accuracy	Macro Precision	Macro Recall	Macro F1
Logistic Regression	0.8977	0.8565	0.8542	0.8547
Voting Ensemble: LR + SVC + SGD	0.8954	0.8532	0.8483	0.8502
Linear SVM	0.8935	0.8491	0.8442	0.8463
HistGradientBoosting on SVD features	0.8852	0.8384	0.8369	0.8370

Random Forest on SVD features	0.8819	0.8425	0.8293	0.8295
Extra Trees on SVD features	0.8819	0.8372	0.8275	0.8286
SGD Linear SVM	0.8234	0.8056	0.7530	0.7475



Among all evaluated models, Logistic Regression achieved the highest overall performance on validation set, obtaining an accuracy of 0,8977 and a macro-F1 score of 0.8547. The Voting Ensemble and Linear SVM produced very similar results, trailing Logistic Regression by less than one percentage point in macro-F1. In contrast, the SGD Linear SVM showed noticeably worse performance, particularly in terms of macro recall, indicating difficulties in consistently identifying all classes.

The relatively small performance gap between Logistic Regression and the Voting Ensemble suggests that the individual linear models make similar prediction errors, limiting the benefit of combining them through majority voting.

5.4. Best evaluated model results

5.4.1 Classification report

Logistic Regression achieved the highest validation macro-F1 among all evaluated classical models and was therefore selected as the representative linear approach for final evaluation on the held-out test set. On the test set, the model achieved an accuracy of 0,8911 and a macro-F1 score of 0,8458, demonstrating that linear classifiers combined with TF-IDF features provide a strong baseline for fine-grained cyberbullying classification.

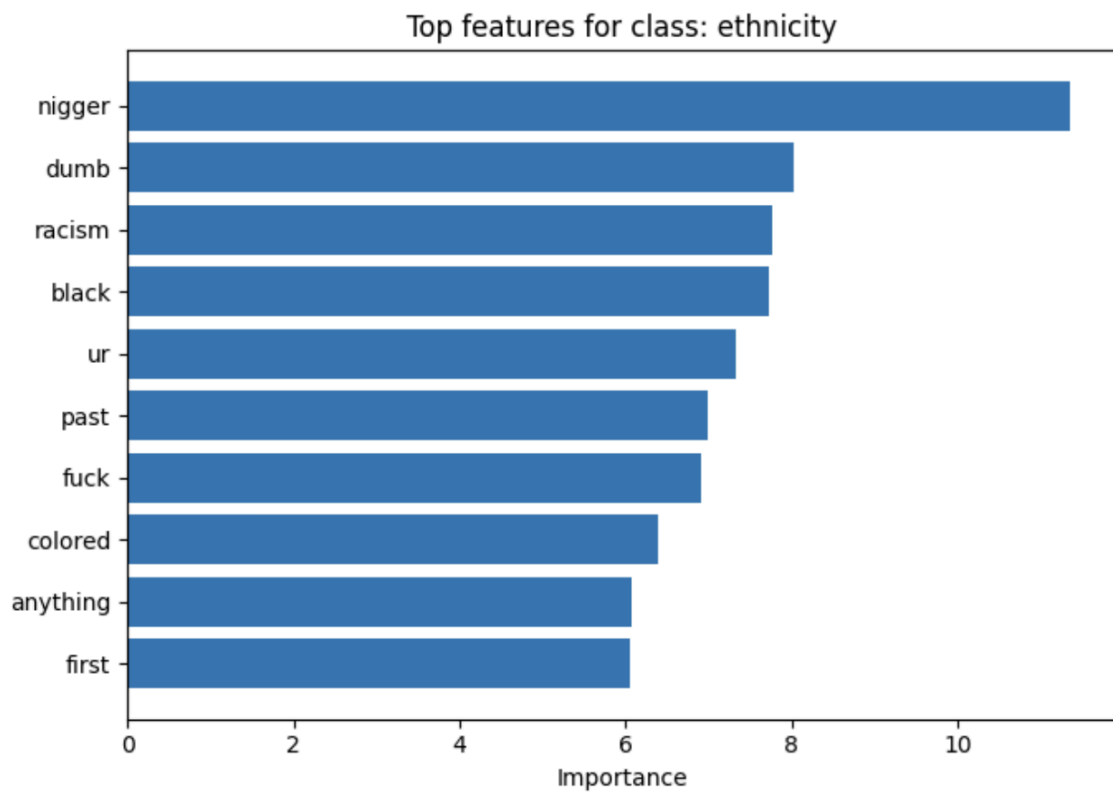
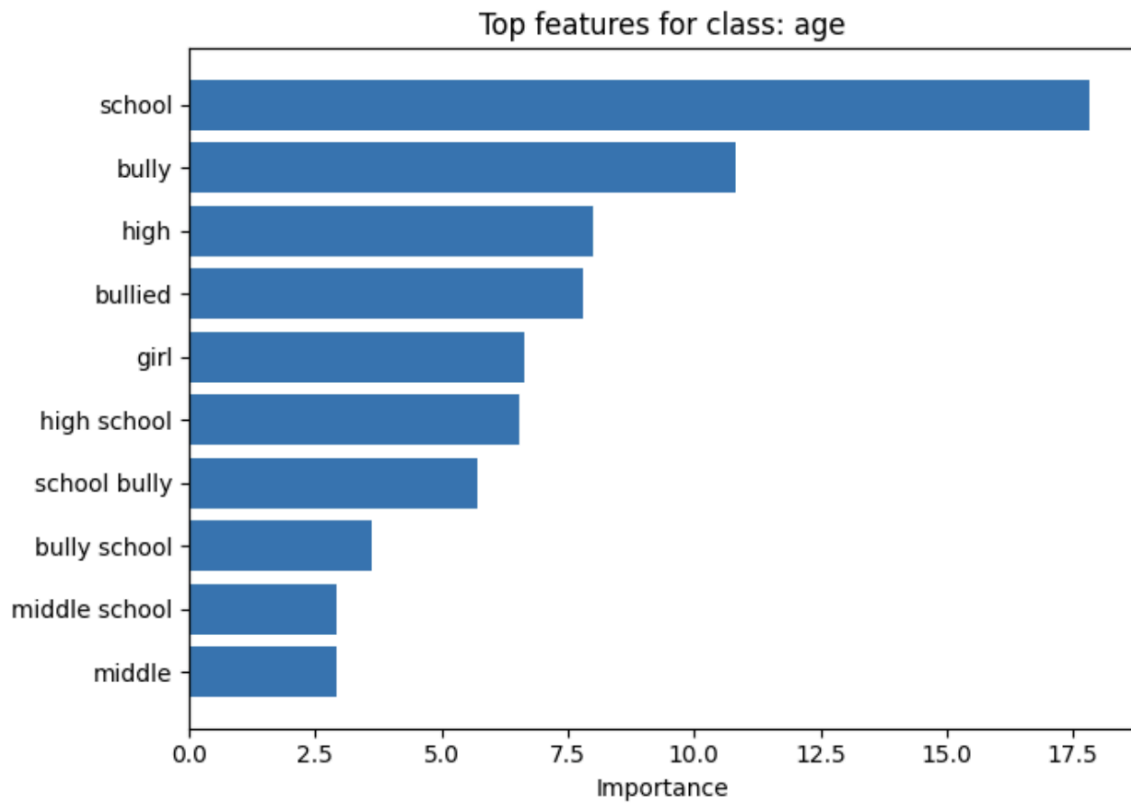
Class	Precision	Recall	F1-score	Support
age	0.9784	0.9874	0.9829	1190
ethnicity	0.9868	0.9853	0.9860	681
gender	0.9276	0.8745	0.9003	733
not_cyberbullying	0.6196	0.5095	0.5592	422
other_cyberbullying	0.6108	0.7511	0.6737	466
religion	0.9727	0.9727	0.9727	732
Macro average	0.8493	0.8467	0.8458	4224
Weighted average	0.8935	0.8911	0.8908	4224

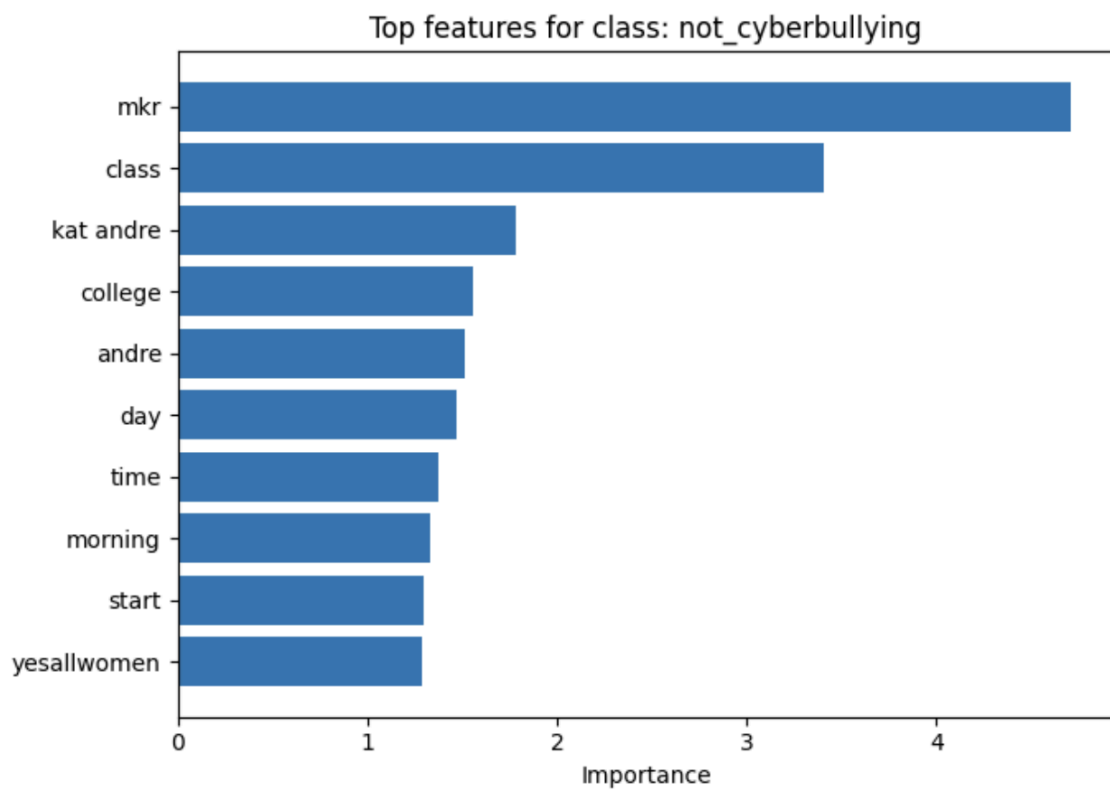
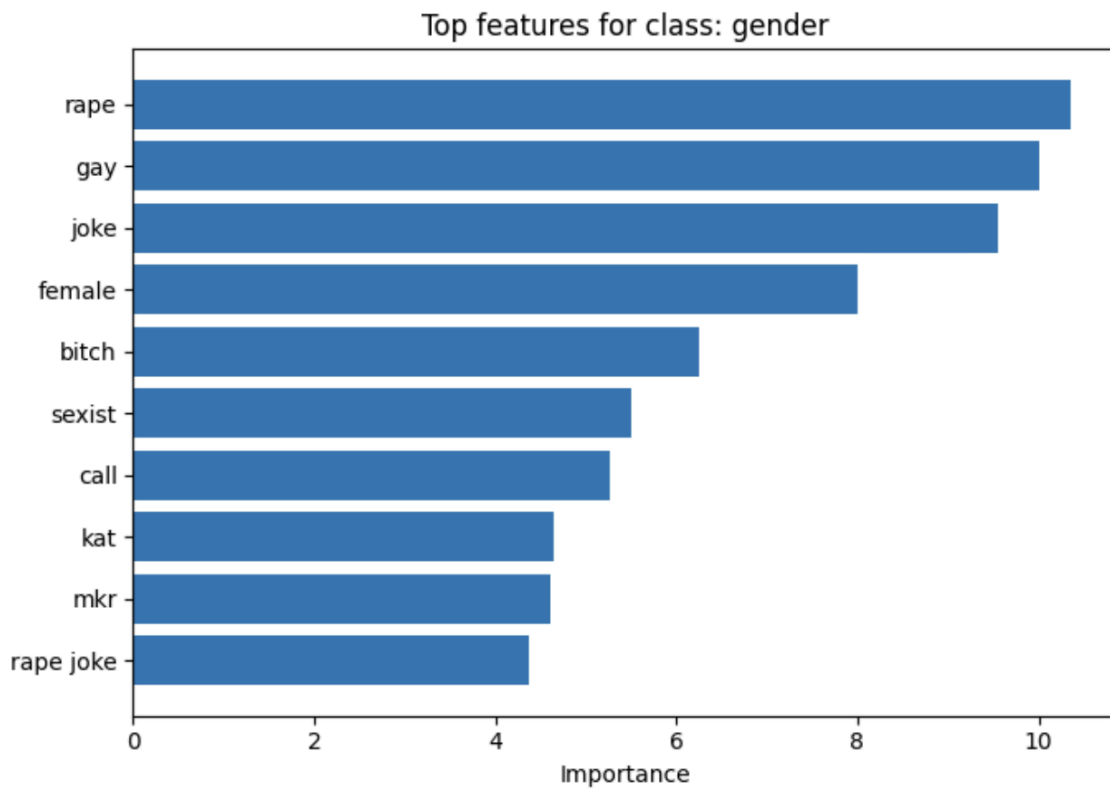
The detailed classification report shows that the model performs exceptionally well on the *age*, *ethnicity*, and *religion* classes, all achieving F1-scores above 0.95. These categories contain highly distinctive vocabulary identified during the exploratory data analysis phase, making them particularly suitable for TF-IDF based classification. Performance on *gender* class is slightly lower but also remains strong.

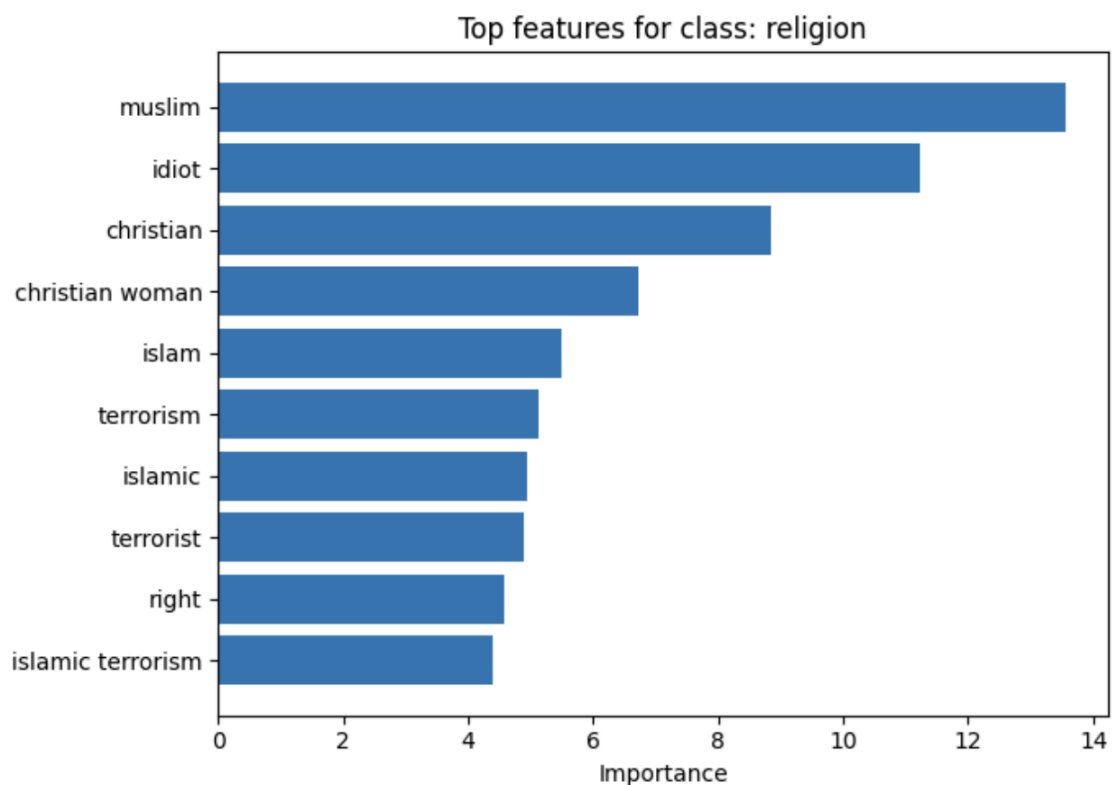
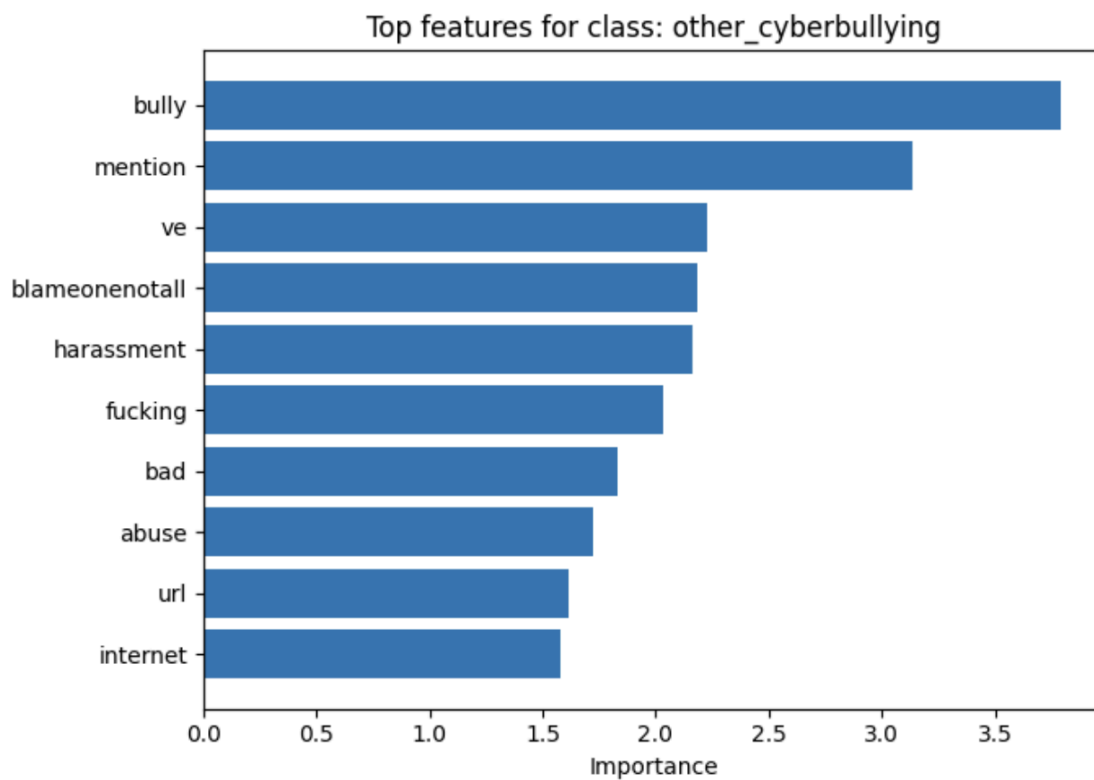
The most challenging classes are *not_cyberbullying* and *other_cyberbullying*, which obtain substantially lower precision and recall scores. This suggests that these categories exhibit considerable semantic overlap and less distinctive lexical patterns, making them inherently more difficult to separate.

5.4.2 Feature importance

One advantage of Logistic Regression is its interpretability through examination of feature coefficients. The terms with the largest positive coefficients for each class represent the features that contribute most strongly to the model assigning a tweet to that category. Each of 6 following figures present the ten most influential features identified by the Logistic Regression classifier for each cyberbullying category:







The extracted features closely align with the patterns observed during exploratory data analysis. The **age** class is dominated by terms related to school bullying experiences, including *school*, *bully*, *bullied*, *high school*, and *middle school*. These features reflect the

autobiographical narratives frequently encountered in this category and explain the model's excellent performance (F1-score is 0.9829).

Similarly, the **ethnicity** class is characterized by explicit racial insults and ethnicity-related vocabulary, such as *nigger*, *racism*, *black*, and *colored*. The highly distinctive nature of these terms contributes to the strong classification performance achieved for this class (F1-score is 0.9860).

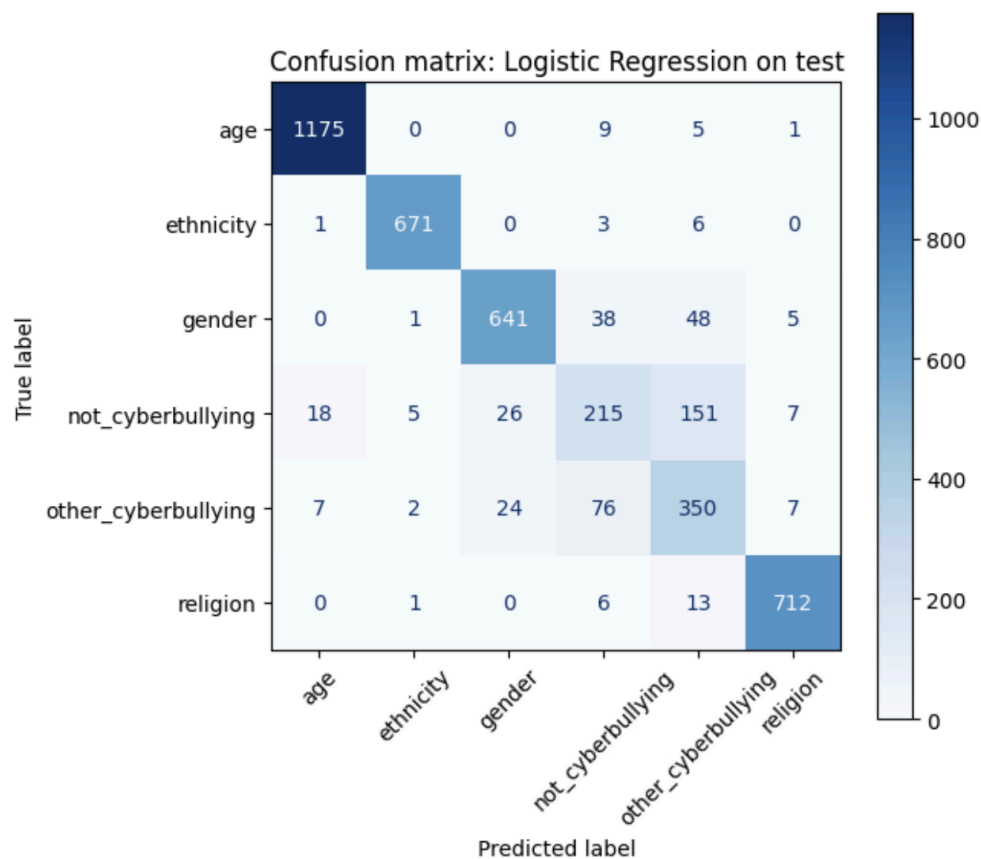
The **gender** category is associated with terms related to sexism and sexual harassment, including *rape*, *gay*, *joke*, *sexist*, *bitch*. These features capture both direct abusive language and common expressions associated with gender-targeted cyberbullying, resulting in a high F1-score of 0.9003.

For the **religion** class, the most important features include *muslim*, *christian*, *islam*, *islamic*, *terrorism*, and *islamic terrorism*. These terms reflect the strong thematic focus of religious cyberbullying content and explain very good classification performance observed for this category (F1 = 0.9727).

In contrast, the **not_cyberbullying** and **other_cyberbullying** classes exhibit less coherent feature sets. The **not_cyberbullying** class is dominated by terms such as *mr*, *class*, *college*, and other context-specific expressions, suggesting that this category is partly influenced by dataset-specific sampling artifacts. The **other_cyberbullying** category contains more generic abusive vocabulary, including *bully*, *harassment*, *abuse*, and *fucking*, together with Twitter-specific terms such as *mention* and *URL*. The absence of highly distinctive lexical markers for these classes likely contributes to their substantially lower classification performance and the frequent confusion observed between them.

The feature importance analysis demonstrates that the Logistic Regression model successfully captures meaningful linguistic patterns associated with different forms of cyberbullying. Furthermore, the learned feature weights closely correspond to the observations made during exploratory data analysis, providing additional evidence that the model relies on interpretable and semantically relevant signals rather than spurious correlations.

5.4.3 Error analysis



The confusion matrix of the Logistic Regression model reveals that the majority of classification errors are concentrated between the *not_cyberbullying* and *other_cyberbullying* classes. In contrast, relatively few errors occur among the four topic-specific categories (age, ethnicity, gender, and religion), indicating that the model successfully captures the distinctive lexical characteristics of these classes.

The strongest source of confusion is the misclassification of *not_cyberbullying* tweets as *other_cyberbullying* and vice versa. This observation is consistent with the vocabulary analysis presented in Section 2.2.3. The *not_cyberbullying* class contains relatively few unique terms and often overlaps lexically with other categories, whereas *other_cyberbullying* represents a heterogeneous residual category encompassing diverse forms of abusive language that do not fit the more specific labels.

Overall, the error patterns suggest that the limitations of the classical approach arise less from insufficient model capacity and more from the inherent ambiguity and overlap between these two categories. The near-perfect performance on the remaining classes demonstrates that TF-IDF representations combined with linear models are highly effective when classes exhibit distinctive lexical signals.

6. Transformer model

The Transformer component fine-tunes DistilBERT (distilbert-base-uncased), a distilled variant of BERT that retains most of BERT's accuracy while being roughly 40% smaller and faster, which makes

fine-tuning feasible on limited hardware. Tweets are short, so a maximum sequence length of 96 subword tokens is used. Only light normalization is applied before tokenization: URLs and mentions are replaced with placeholder tokens. Unlike the classical pipeline, stopword removal and lemmatization are not applied, since the Transformer is designed to use the contextual and function-word information that such preprocessing discards. The model is trained on the same deduplicated dataset as the linear and ensemble models, selected on the validation split by macro-F1, and evaluated once on the held-out test split. The two parts share the identical test rows, so all reported scores are directly comparable.

6.1. Transformer training configuration

DistilBERT was fine-tuned for 2 epochs with a learning rate of $2e-5$, a training batch size of 16, and a cross-entropy objective over the six encoded labels. The checkpoint with the best validation macro-F1 was retained for the final test evaluation.

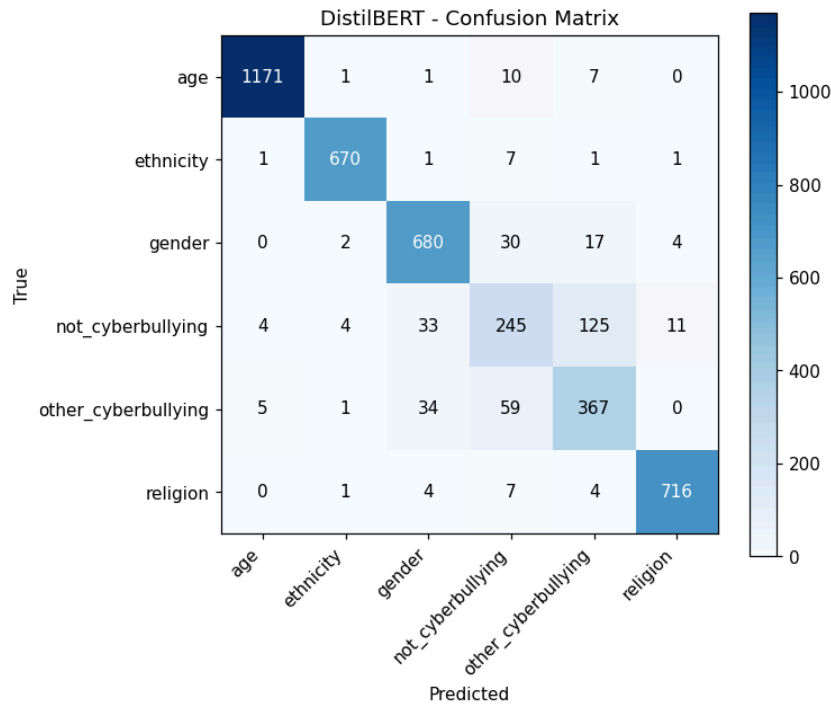
6.2. Transformer results

On the held-out test set of 4,224 tweets, the fine-tuned DistilBERT achieves an overall accuracy of 0.9112 and a macro-F1 of 0.8730 (macro precision 0.8747, macro recall 0.8736). The per-class breakdown is given below.

Class	Precision	Recall	F1-score	Support
age	0.9915	0.9840	0.9878	1,190
ethnicity	0.9867	0.9838	0.9853	681
gender	0.9031	0.9277	0.9152	733
not_cyberbullying	0.6844	0.5806	0.6282	422
other_cyberbullying	0.7044	0.7876	0.7437	466
religion	0.9781	0.9781	0.9781	732
Macro avg	0.8747	0.8736	0.8730	4,224

Table. Per-class performance of fine-tuned DistilBERT on the test set.

6.3. Error analysis



confusion matrix

Figure. Confusion matrix for fine-tuned DistilBERT on the test set (rows = true class, columns = predicted class).

The four topic-defined classes — age, ethnicity, religion, and gender — are classified with high reliability (F1 between 0.915 and 0.988). As with the classical model, the overwhelming majority of errors falls in a single block: not_cyberbullying and other_cyberbullying are confused with each other. Of the 422 true not_cyberbullying tweets, 125 are predicted as other_cyberbullying; of the 466 true other_cyberbullying tweets, 59 are predicted as not_cyberbullying. This is consistent with the vocabulary analysis in Section 2.2.3: the four topical classes carry strong distinctive lexical signals, whereas not_cyberbullying has few unique words and other_cyberbullying lacks a coherent thematic vocabulary, so the boundary between them is inherently ambiguous rather than a modeling deficiency.

7. Discussion

7.1. Models comparison

The two model families were evaluated on the identical deduplicated test set of 4,224 tweets with the same metrics. The best classical model, Logistic Regression on the TF-IDF and meta-feature representation, reaches an accuracy of 0.8911 and a macro-F1 of 0.8458. The fine-tuned DistilBERT reaches an accuracy of 0.9112 and a macro-F1 of 0.8730, improving macro-F1 by 2.7 points over the linear model.

Model	Accuracy	Macro precision	Macro recall	Macro F1
Logistic Regression	0.8911	0.8493	0.8467	0.8458
DistilBERT	0.9112	0.8747	0.8736	0.8730

Table. Linear/ensemble vs Transformer on the shared test set.

Both models show the same qualitative pattern: near-perfect performance on the four topical classes, where distinctive vocabulary already makes the bag-of-words signal highly discriminative, and a sharp drop on the `not_cyberbullying` and `other_cyberbullying` classes. The improvement from the Transformer is concentrated almost entirely in this difficult block: the F1-score for `not_cyberbullying` rises from 0.5592 to 0.6282 and for `other_cyberbullying` from 0.6737 to 0.7437, while the four topical classes are already near-saturated for both models and change little. This is the expected benefit of contextual language modeling: where surface n-grams are insufficient to separate generic non-cyberbullying from diffuse “other” cyberbullying, DistilBERT’s contextual representations recover part of the signal. At the same time, this boundary remains the dominant source of error for both models, indicating that the residual confusion reflects genuine semantic overlap in the label definitions rather than model capacity alone.

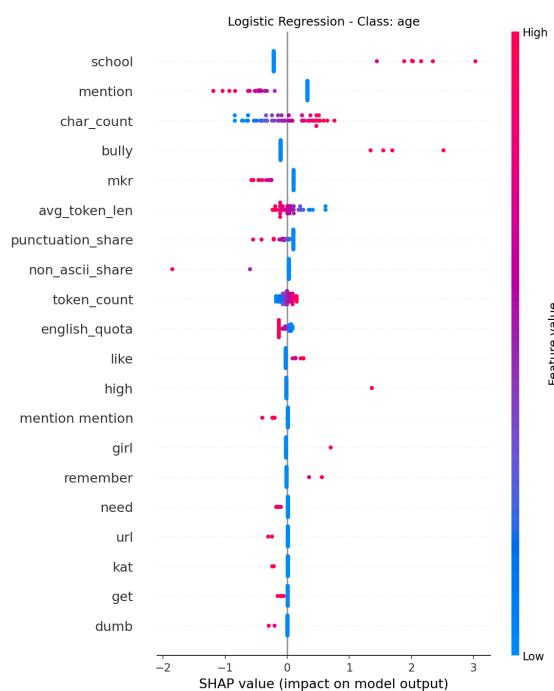
7.2. SHAP analysis

For the SHAP analysis only those models that have `predict_proba` method have been analysed through the *shap* (SHapley Additive Explanations) library. In addition, it was decided to leave Extra Trees and Random Forest as they require additional preprocessing for the production of SHAP plots and they would not contain any meaningful information. Thus, only logit model is analysed

For each predicted class the SHAP diagram was plotted:

- Age

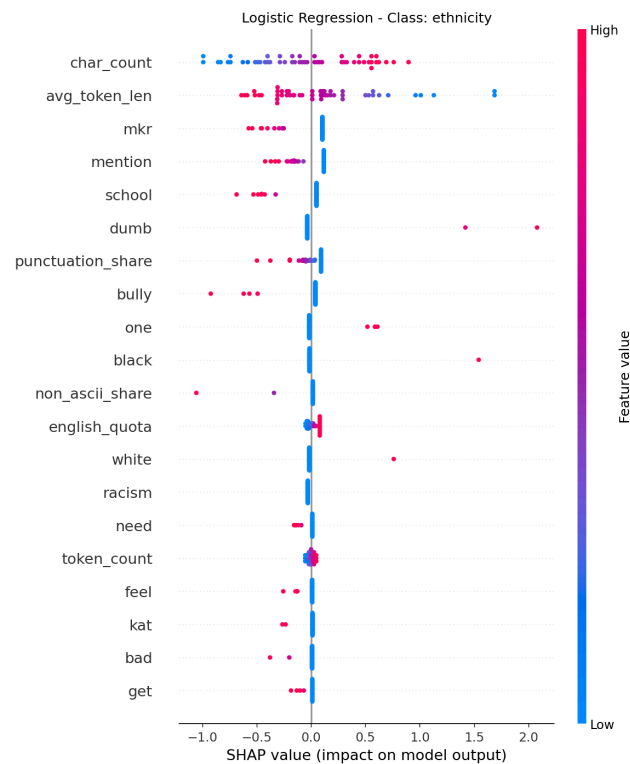
Top features pushing predictions toward age-related bullying: `school`, `mention` (tagging others), `char_count` (longer tweets), `bully`, etc. Age-related cyberbullying often occurs in school contexts, involves tagging victims, and uses longer, more descriptive tweets.



Another interesting finding is the “mkr” text which often appears for various cyberbullying types. Standing for “My Kitchen Rules” and often used as a hashtag, it is a great challenge for linear models to separate it based on the context (which transformers can do).

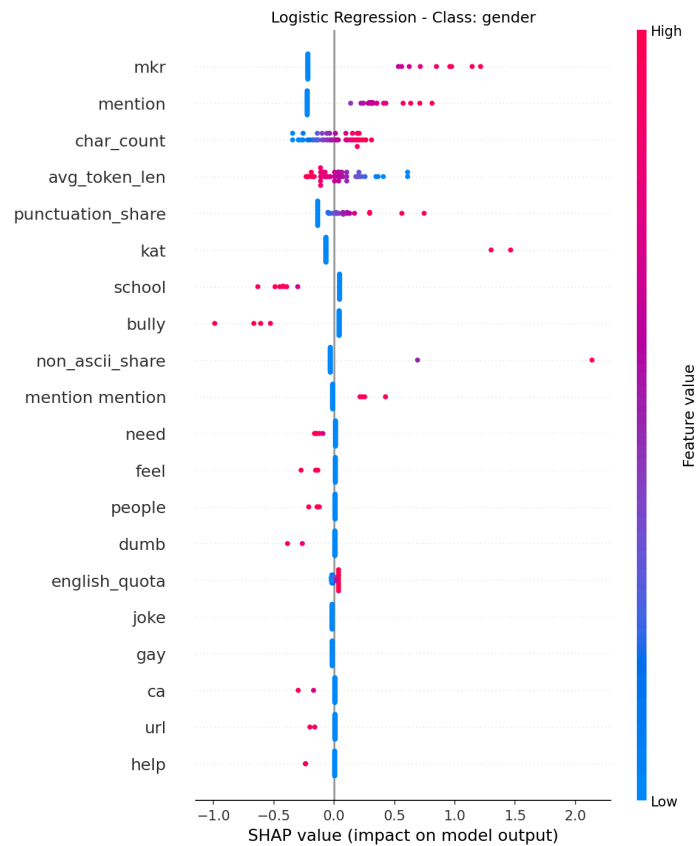
- Ethnicity

Top features predictions toward ethnicity-related bullying: dumb, black/white (direct racial references), racism, people (general group references e.g. “these people ...”). Ethnicity bullying explicitly uses racial terminology (black, white, racism) and offensive terms (dumb).



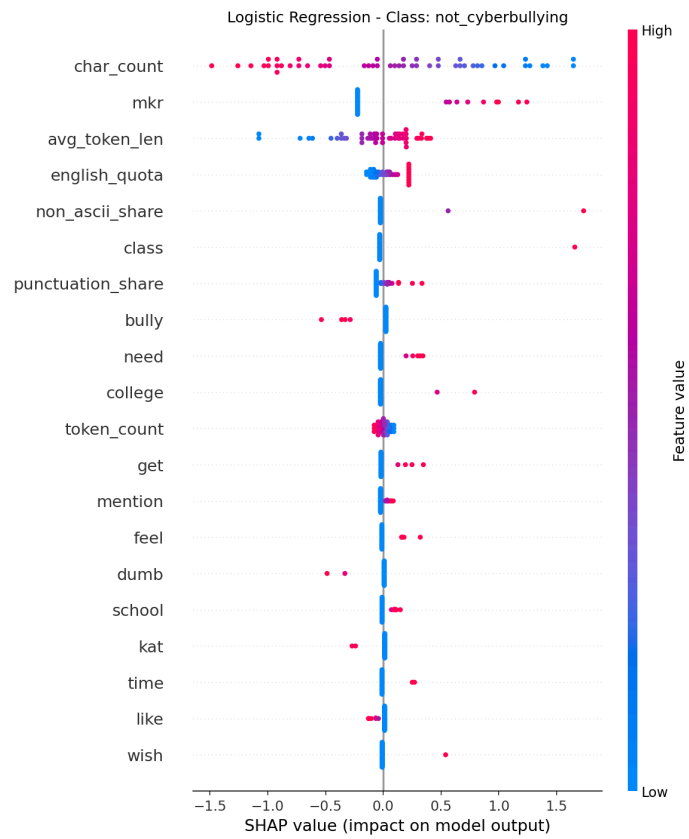
- Gender

Top features pushing predictions toward gender-related bullying: joke (bullying framed as humor), direct sexual orientation, mentioning. Gender-related cyberbullying often uses sexual orientation slurs and frames bullying as "jokes" to minimize harm.



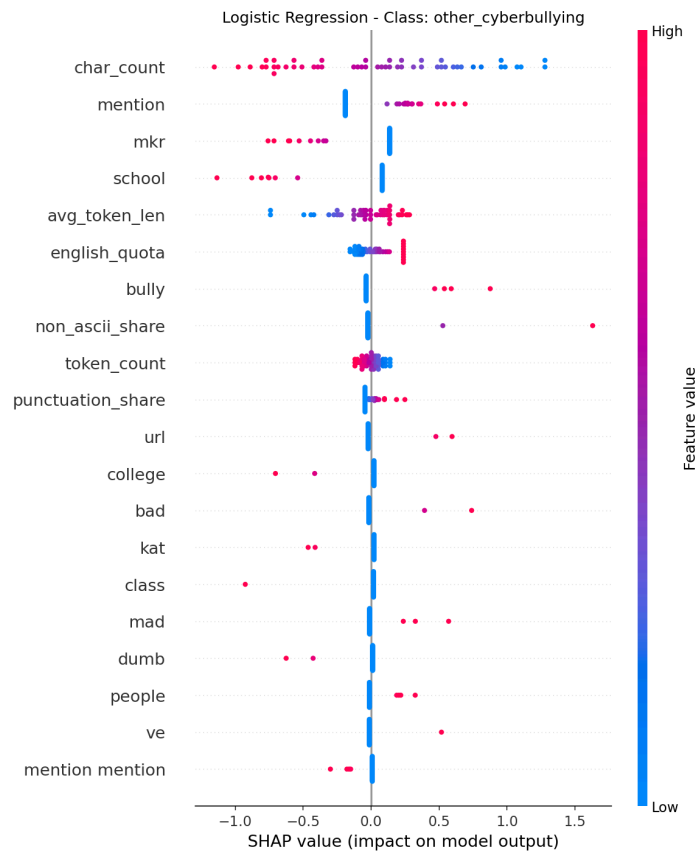
- Not cyberbullying

Features pushing predictions toward not cyberbullying: char_count (shorter tweets), greater percentage of english words, more punctuation, more special characters (non-ascii). Non-bullying tweets tend to be shorter, lack bullying vocabulary, and don't express urgent needs.



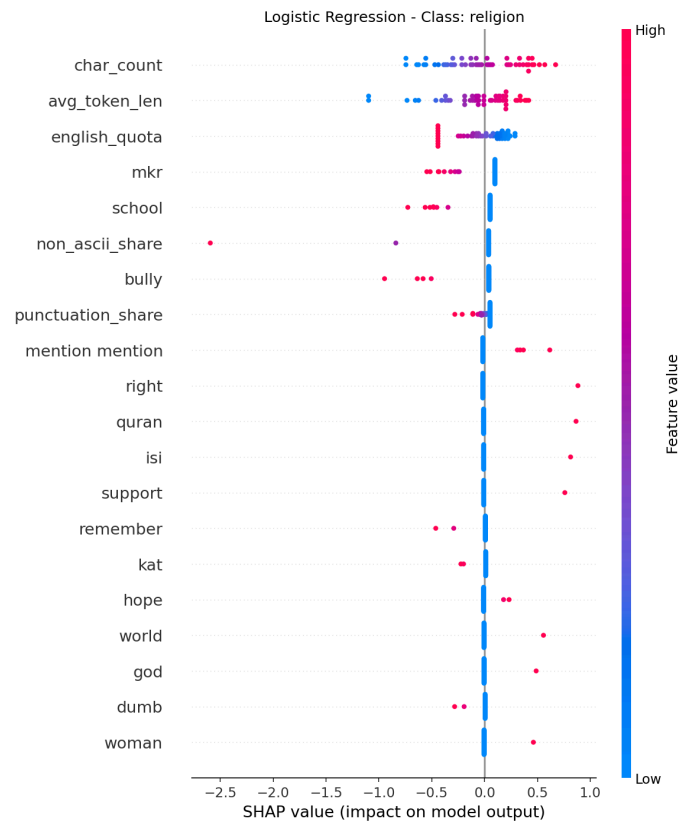
- Other cyberbullying

Exhibits similar characteristics to already described types. This category is characterized by general tweet features rather than specific vocabulary, suggesting it captures diverse bullying types.



- Religion

Features pushing predictions toward religion-related bullying: quran (religious text reference), isi (possibly religious extremist terrorists), god and other similar to previous characteristics. Religious cyberbullying has very distinctive vocabulary (quran, god, isi) that other classes don't share, making it the most identifiable class.



8. Conclusion

Two complementary approaches were applied to fine-grained six-class cyberbullying classification and evaluated on the identical deduplicated test set: classical linear and ensemble models on a TF-IDF and meta-feature representation, and a fine-tuned DistilBERT Transformer on the raw text. The best classical model (Logistic Regression) reached 0.8911 accuracy and 0.8458 macro-F1; the Transformer reached 0.9112 accuracy and 0.8730 macro-F1. The Transformer's advantage comes almost entirely from better separation of the two semantically diffuse classes (not_cyberbullying and other_cyberbullying), while both models classify the four topically distinct categories almost perfectly. The remaining error is concentrated in the same not_cyberbullying / other_cyberbullying overlap for both approaches, which mirrors the vocabulary analysis from the exploratory phase and points to label ambiguity rather than model capacity as the limiting factor. The shared deduplicated split and common metric set make this comparison valid on equal footing.

GitHub Page - https://github.com/RomanGorelsky/MMDA_Project